

1. How many sequences are there for the first-row Seats in the classroom, if we only consider majors (Biology vs Geography)? Assume 40 Biology students and 30 Geography students in the class and 5 seats available.

2. Based on experts' analysis, Silver State Fighters have winning probability 60% against Boston Green Sox. In the best of seven Finals, what the probability that the series ends at exactly 6 games?

3. We have a card game with 52 cards, consisting of 4 suits and the cards are numbered Ace, 2-10, then J, Q, K in each suit. A hand has 5 cards and can be sorted anyway the player desires.

Four of a kind: AAAA9

Full House: AAKKK, 22333

Two Pair (Distinct):

4. If we randomly draw two cards from a deck of 52 cards without replacement. Let A: the first card is K and B: the second card is K. Are events A and B independent? Explain why.

5. Suppose 7% of all major league baseball players use an illegal steroid. A league test is considered 96% effective in detecting the illegal steroid, but unfortunately, it also incorrectly tests positive (indicating steroid use) for a person who does not have the illegal steroid in their system 12% of the time.

1. What proportion of the league players will test positive for the illegal steroid?
2. Given that someone tests positive for an illegal steroid, what is the probability that they do not have an illegal steroid in their system?
3. Suppose four players have taken steroids. What is the probability all of them are detected by the test?

6. Let the random variable X denote the number of wafers that need to be analyzed to detect a (FIRST) large particle of contamination. Assume that the probability that a wafer contains a large particle is 0.01, and that the wafers are independent. Determine the type of probability distribution of X .

7. A day's production of 850 parts contains 50 defective parts. Two parts are selected at random without replacement. Let the random variable X equal the number of defective parts in the sample. Find the cumulative distribution function of X .

8. Roll a dice or a die once. Let X be the random number defined by the number of the dice. Determine $E[X]$, $V(X)$, and the standard deviation.

9. Assume a lot contains 20 wafers that each has a defect rate of 30%. Find the probabilities of the following situations:

1. Exactly 3 wafers are defective.
2. At most 5 wafers are defective.
3. More than 10 wafers are defective.
4. More than 2 but less than 5 wafers are defective.

10. Based on previous experience, the mean number of customer service calls during a 3-hour period is 300 at Amazon. Answer the following questions:

1. What is the probability of exactly 40 calls in 30 minutes?
2. What is the probability of at most 120 calls in an hour?
3. What is the probability of more than 10 calls in 10 minutes?

11. Suppose the time that a student of ISyE6739 takes to prepare for the Midterm is uniformly distributed between 1 to 10 days.

1. What's the probability that a student spends less than 3 days preparing for the Midterm?
2. What's the average number of days that a student spends for the Midterm?
3. What's the probability that a student spends less than 6 days but more than 2 days preparing for the Midterm?

12. Assume X is normally distributed with a mean of 5 and a standard deviation of 4. Determine the following:

- a. $P(X < 11)$
- b. $P(X > 0)$
- c. $P(3 < X < 7)$

13. The compressive strength of samples of cement can be modeled by a normal distribution with a mean of 6000 kilograms per square centimeter and a standard deviation of 100 kilograms per square centimeter.

- (a) What is the probability that a sample's strength is less than 6250 Kg/cm²?
- (b) What strength is exceeded by 95% of the samples?

14. In a large corporate computer network, user log-ons to the system can be modeled as a Poisson process with a mean of 25 log-ons per hour.

- 1. What is the probability that there are no log-ons in the next 6 minutes?
- 2. What is the probability that the time until the next log-on is between 2 and 3 minutes?
- 3. What is the interval of time such that the probability that no log-on occurs during the interval is 0.90?

15. Assume McDonald's want to understand the usage of drive-thru. Assume we randomly pick one day and let X and Y denote the usages of drive-thru @ Northside Dr. and Howell Mill Rd., respectively. Assume the joint PDF of X and Y is as follows:

$$f_{XY}(x, y) = \begin{cases} \frac{2}{3}(x + 2y), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Find $f_X(x)$ and $f_Y(y)$ then determine the probability that the usage @ Northside Dr. is above 50%.

16. Let Y denote the fuel (in 1000 gallons) stored at a local gas station for a day. Let X denote the number of gallons sold in a day (in 1000 gallons). Assume the fuel tank capacity is 1000 gallons. The joint pdf is:

$$f_{XY}(x, y) = 2, 0 < x < y, 0 < y < 1$$

1. Find $f_X(x)$.
2. What is $P(0.25 < X < 0.5 | Y = 0.75)$
3. Find $E(X | y)$.

17. The service of Ethan's Car Wash includes three major steps – rinse, dry, and wax. Suppose that the times of these three steps are normally distributed and mutually independent with means 10, 5, 15 minutes and variance 4, 3, 9 respectively. Answer that following questions:

- (a) What is the mean of the total time for washing a car at Ethan's Car Wash?
- (b) What is the standard deviation of the total time for washing a car at Ethan's Car Wash?
- (c) What is the probability that the total time for washing a car is less than 40 minutes?
- (d) What is the probability that the total time for washing a car is greater than 25 minutes?

1. 32
2. 0.29952;
3. 424; 3744; 123552
4. No

5. 0.1788; 0.62416; 0.8493
6. Geometric Distribution

7.
$$F(x) = \begin{cases} 0, & \text{if } x < 0 \\ 0.885748, & \text{if } 0 \leq x < 1 \\ 0.996605, & \text{if } 1 \leq x < 2 \\ 1, & \text{if } x \geq 2 \end{cases}$$

8. 3.5; 2.9167
9. 0.07160; 0.4164; 0.0171; 0.2020
10. 0.0215; 0.9773; 0.9427
11. 0.2222; 5.5; 0.4444
12. 0.9332; 0.8944; 0.3829
13. 0.9938; 5835.5
14. 0.0821; 0.1481; 0.2529 minutes
15. 0.5833
16. $f_{\{X|y\}}(x)=1/y$ and $f_Y(y)=2y$; 0.3333; $y/2$
17. 30; 4; 0.9938; 0.8944